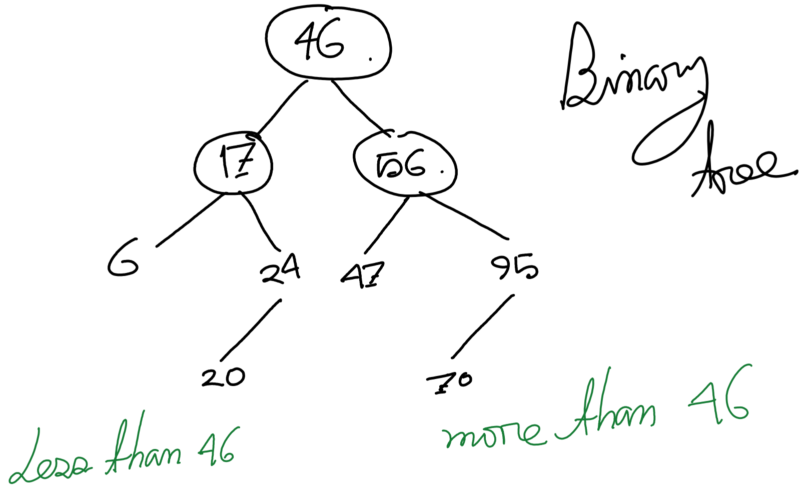
Binary Search Tree



Properties of Binary Search Tree:

1. The left subtree of a node contains only nodes with keys lesser than the node's key.
2. The right subtree of a node contains only nodes with keys greater than the node's key.
3. The left and right subtree each must also be a binary search tree.
4. No value can be repeated in a binary search tree because the values are key values.

Purpose of Binary Search Tree:

1. Faster searching

   Example: find item = 100

   a. The 100 is more than 46. Therefore it will not be in the left subtree
   b. Now in the right subtree, the node is 56. Therefore the value 100 should be on the right subtree of node 56
   c. Now 95 is less than 100. Therefore we will go to the right subtree. However, there is no right subtree
   d. That means the 100 is not in the binary search tree

Depth of the Binary Search Tree:

1. From root node to the leaf node, maximum number of node is the depth of a binary search tree